Understanding Tree Model Dominance in Tabular Learning - Performance, Robustness, and Stability Across Classical, Deep and Foundation-Style Models

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***Abstract*—Tree-based models remain highly competitive on tabular learning tasks despite recent progress in deep and pretrained tabular models. This paper presents an exploratory empirical comparison of classical tree ensembles, neural tabular models, and pretrained tabular models across six classification and regression datasets. Beyond clean-data performance, we evaluate robustness under controlled missingness and synthetic high-cardinality stress, and we analyze covariate-shift behavior using predictive degradation and SHAP-based feature-stability measures.**

Across the selected benchmark, boosted tree methods show the most consistent empirical pattern: they usually remain close to the best clean-data performance while also preserving stronger robustness and feature-attribution stability than the neural baselines. Neural and pretrained models are competitive in selected cases, but their behavior is less consistent across perturbation, shift, and attribution-stability criteria. These findings should be interpreted as evidence from a controlled benchmark rather than as a universal claim. Overall, the results suggest that the strength of tree-based models in this study is linked not only to predictive accuracy, but also to their stability under controlled tabular stress conditions.

***Index Terms*—tabular machine learning, tree-based models, deep learning, hyperparameter optimization, robustness, covariate shift, SHAP, feature attribution stability**

I. Introduction

I-A. Background and Motivation

Tabular data is one of the most common data formats in applied machine learning and appears in domains such as finance, healthcare, logistics, and industrial analytics. Although deep learning has achieved major success in computer vision and natural language processing, tree-based methods remain strong baselines for many tabular prediction tasks. In practice, models such as Random Forest, XGBoost, LightGBM, and CatBoost are often difficult to outperform, especially on datasets with mixed feature types, moderate sample sizes, and irregular feature–target relationships.

This continued strength raises an important question: why do tree-based models remain so effective on many tabular datasets? A comparison based only on clean benchmark performance is not sufficient to answer this question. In practical applications, tabular models may also face incomplete inputs, high-cardinality categorical variables, and changes in the test-time feature distribution. In this study, we approximate these issues through controlled stress tests and covariate-shift perturbations. These experiments are intended as heuristic probes of model behavior, not as exhaustive simulations of all real deployment conditions.

The goal of this paper is therefore to provide an exploratory empirical comparison of classical tree ensembles, neural tabular models, and recent pretrained tabular models under multiple evaluation views. We compare clean hyperparameter-optimized performance, robustness to missingness and synthetic high-cardinality nuisance features, and SHAP-based feature-attribution stability under covariate shift. By combining these perspectives, the study aims to identify recurring reliability patterns across the selected benchmark rather than to make universal claims about all tabular learning problems.

II. Related Work

Tree-based ensembles remain the dominant classical baseline for supervised tabular learning. Random Forests introduced bagging-based tree ensembles as a robust nonparametric baseline [1], while XGBoost established a scalable and highly effective gradient-boosting framework [2]. LightGBM further improved the efficiency of gradient boosting through histogram-based learning and leaf-wise tree growth [3], and CatBoost strengthened tabular performance through ordered boosting and explicit handling of categorical features [4]. These models remain central reference points because they combine strong predictive accuracy with practical efficiency and comparatively stable behavior on many structured datasets.

A large body of work has attempted to close this gap with neural architectures specialized for tabular data. TabNet introduced sequential attentive feature selection as an interpretable deep alternative [5], while TabTransformer and SAINT extended this direction through attention-based contextualization and contrastive pretraining [6], [7]. Gorishniy et al. later showed that carefully tuned simple baselines, especially ResNet-style models and FT-Transformer, are substantially stronger than many earlier bespoke tabular neural networks, but still do not consistently displace boosted tree ensembles [8].

Recent benchmark efforts have broadened this discussion. McElfresh et al. introduced TabZilla, a large-scale benchmark showing that the relative performance of neural and tree-based methods depends strongly on dataset characteristics and evaluation design [9]. Erickson et al. introduced *TabArena*, a continuously maintained benchmark for tabular machine learning with protocol-aware evaluation and public leaderboards [10]. The study most directly related to our interpretation is that of Grinsztajn et al., who argue that tree-based models remain strong on typical tabular data because their inductive biases are better aligned with uninformative features, feature orientation, and irregular decision structure [11].

More recently, pretrained and zero-shot tabular models have shifted part of the comparison. TabPFN frames tabular prediction as in-context inference using a transformer pretrained on synthetic tasks, giving strong performance especially on small classification problems [12]. APT extends this direction through adversarially pretrained tabular prediction [13]. These models reduce part of the dataset-specific tuning burden, but their robustness and feature-stability behavior under perturbation and shift remain less established than their clean benchmark performance.

Robustness-oriented tabular evaluation has also received increasing attention. TableShift provides a benchmark for distribution shift in tabular data across several real-world domains, showing that in-distribution and out-of-distribution performance can diverge substantially [14]. Gardner et al. further show that tree-based methods are strong baselines for subgroup robustness, often competing with or outperforming specialized robustness and fairness methods [15]. TabularBench studies adversarial robustness for tabular deep learning, highlighting that robustness in structured data requires different assumptions from image-based adversarial settings [16]. Compared with these works, our study focuses on a complementary setting: controlled missingness and high-cardinality stress, covariate-shift evaluation, and feature-attribution stability within one tree-versus-neural-versus-pretrained comparison.

Our stability analysis builds on SHAP, which explains model predictions through additive feature attributions and is widely used to summarize feature importance in tabular models [17]. In this paper, SHAP is not treated as causal evidence; rather, it is used as a diagnostic tool for comparing whether models preserve similar feature-importance rankings under perturbation and covariate shift. For rank-based model comparison, we use the Friedman test and Nemenyi-style post-hoc comparison following the standard classifier-comparison methodology of Demšar [18]. Hyperparameter optimization is performed with Optuna, which provides an efficient framework for automated search across heterogeneous model families [19]. The datasets used in the benchmark are drawn from commonly used public tabular sources, including Titanic, Telco Churn, DryBean, the scikit-learn digits dataset, and Diamonds [20]–[24].

TABLE I: Summary of datasets used in this study.

Dataset	Task	Samples	Feat.	Num/Cat	Note
diamonds_v2	Reg.	219,703	24	6/18	Pearson 0.85
diamonds_v3	Reg.	219,703	22	4/18	Pearson 0.70
drybean	Multi-cls.	4,970	12	12/0	Bean benchmark
mnist	Multi-cls.	1,797	64	64/0	Tabular digits
telco	Binary cls.	7,032	20	4/16	Churn dataset
titanic	Binary cls.	183	11	6/5	Survival task

III. Experimental Setup

III-A. Datasets

We selected six tabular datasets spanning both classification and regression tasks in order to evaluate model behavior across different feature structures, target types, and levels of learning difficulty. The classification benchmark consists of the *Titanic*, *Telco Churn*, *DryBean*, and tabularized *MNIST* datasets, while the regression benchmark consists of two curated variants of the Diamonds dataset: *diamonds_v2* and *diamonds_v3*.

The benchmark is intentionally heterogeneous but moderate in scale. It is therefore not designed to isolate dataset-size effects systematically. In particular, the selected datasets include both very small settings, such as Titanic, and larger settings, such as Diamonds, but they do not provide a controlled learning-curve study across sample sizes. This matters because neural tabular models, including TabNet, may require more data to realize their representational advantages. We therefore interpret dataset-size effects cautiously and treat a systematic subsampling or learning-curve analysis as an important direction for future work.

Each dataset also plays a distinct structural role in the benchmark. *Titanic* represents a small mixed-feature binary classification problem in which limited sample size and heterogeneous attributes can favor models with strong inductive bias. *Telco Churn* provides a medium-scale binary classification task with many categorical predictors, making it particularly relevant for studying categorical handling and sensitivity to nuisance structure. *DryBean* is a multiclass benchmark with primarily numerical shape descriptors, allowing us to assess model behavior in a structured geometric setting. Tabularized *MNIST* serves as a high-dimensional numerical classification task, where the advantages of deep models may become more pronounced than in smaller mixed-type datasets (The file used in this project is not the original 70k-sample MNIST dataset. It is the full 8×8 tabular digits benchmark, which naturally contains 1,797 samples and 64 features).

The two Diamonds variants were included to study regression behavior under different levels of feature redundancy control. Both are derived from the same source dataset, but they differ in the correlation threshold used during feature filtering. In *diamonds_v2*, features were filtered using a Pearson correlation threshold of 0.85, whereas *diamonds_v3* used a stricter threshold of 0.70, removing more moderately correlated predictors. This distinction allows us to assess whether stronger decorrelation affects model behavior differently, particularly for neural models that may respond more sensitively to changes in feature redundancy.

A summary of the datasets, including task type, number of samples, and feature composition, is provided in Table I.

III-B. Models

We evaluated three groups of tabular learning models: *tree-based methods*, *neural tabular models*, and *pretrained tabular foundation-style models*.

The tree-based group includes *Random Forest (RF)*, *XGBoost*, *LightGBM*, and *CatBoost*. These models were selected because they represent the strongest and most widely used classical approaches for tabular prediction. Random Forest serves as a robust bagging-based ensemble baseline, while XGBoost, LightGBM, and CatBoost provide modern gradient-boosted tree frameworks with different design choices for efficiency, split construction, and categorical feature handling.

The neural group includes a standard *multilayer perceptron (MLP)* and *TabNet*. The MLP was included as a simple dense neural baseline, whereas TabNet was chosen as a tabular-specific neural architecture based on sequential attention and feature selection. These two models allow comparison between a general-purpose neural approach and a specialized deep tabular model.

In addition, we included two recent pretrained tabular models, *TabPFN* and *APT*. Unlike the other models, these methods are designed for zero-shot or low-tuning inference using pretrained representations rather than extensive dataset-specific optimization.

A compact summary of the evaluated models is given in Table II.

TABLE II: Summary of evaluated model families and tuning strategy.

Model	Family	Task	HPO	Note
RF	Tree	Cls./Reg.	Yes	Bagging ensemble
XGBoost	Tree	Cls./Reg.	Yes	Gradient boosting
LightGBM	Tree	Cls./Reg.	Yes	Histogram boosting
CatBoost	Tree	Cls./Reg.	Yes	Native categorical support
MLP	Neural	Cls./Reg.	Yes	Dense feed-forward network
TabNet	Neural	Cls./Reg.	Yes	Sequential attention model
TabPFN	Pretrained	Cls.	No	Default pretrained config
APT	Pretrained	Cls.	No	Fixed pretrained checkpoint

This creates an intentional but important asymmetry in the comparison. RF, XGBoost, LightGBM, CatBoost, MLP, and TabNet are dataset-specifically tuned, whereas TabPFN and APT are evaluated as pretrained classification models using their available default or fixed-checkpoint configurations. Consequently, the pretrained-model results should not be interpreted as a fully symmetric HPO comparison. Instead, they indicate how these models behave in the practical low-tuning setting for which they are commonly used. Because TabPFN and APT are classification-only in this study, they are excluded from regression summaries.

III-C. Hyperparameter Optimization Protocol

For the tunable models, hyperparameter optimization was performed separately for each dataset–model pair using Optuna [19]. Each optimization run used 30 trials, and the best configuration was selected according to validation performance before final evaluation on the held-out test set. Classification models were selected using validation F1-score, while regression models were selected using validation R^2.

The data were split into training, validation, and test partitions before model fitting. Preprocessing objects, including imputers, encoders, and scalers where applicable, were fitted only on the training data and then applied consistently to validation and test data. The validation split was used only for hyperparameter selection, while the test split was held out until final evaluation. Random seeds were fixed for the main experimental runs to improve reproducibility.

Optuna was chosen because it provides an efficient and flexible search framework for heterogeneous model families with mixed continuous, discrete, and categorical hyperparameters. For tree-based models, the search spaces included parameters such as the number of trees or iterations, maximum depth, learning rate, subsampling, column sampling, minimum child constraints, leaf parameters, and regularization terms depending on the model family. For neural models, the search spaces included architectural and optimization parameters such as number of layers, hidden dimensions, dropout, learning rate, batch size, and early-stopping patience. For TabNet, model-specific parameters such as decision dimension, attention dimension, number of decision steps, sparsity regularization, and optimizer learning rate were also included.

The same trial budget was used for the tunable model families to keep the experimental protocol simple and comparable. However, equal trial counts do not guarantee equal search fairness because model families differ in the number, type, and sensitivity of their hyperparameters. Neural models and boosted trees can require different effective tuning budgets, and the 30-trial budget should therefore be interpreted as a bounded practical HPO setting rather than an exhaustive search. We explicitly account for this limitation when interpreting model-family differences.

The full search-space ranges, preprocessing settings, and random seeds are provided in the released experiment configuration files. The main reported runs used random seed 42 for data splitting and model initialization where supported.

III-D. Robustness Stress Scenarios

We evaluated robustness under two controlled stress scenarios: missingness and high-cardinality categorical noise. These experiments were applied to diamonds v2, Telco, DryBean, and tabularized MNIST, covering regression, binary classification, multiclass classification, mixed numerical–categorical inputs, and purely numerical feature spaces. Titanic was excluded because of its small sample size and existing missingness, while diamonds v3 was excluded because it is a structural variant of diamonds v2.

The perturbation levels were chosen as moderate heuristic stress tests rather than domain-calibrated simulations. Their purpose is to expose model-family differences under controlled feature-space degradation, not to reproduce a specific deployment process.

For the missingness scenario, we randomly selected 20% of feature columns and removed 15% of values within those columns in both the training and test sets. Models were then trained and evaluated under matched missingness conditions. Numerical missing values were imputed before model fitting.

Categorical missing values were represented using a dedicated "Missing" level where possible; otherwise, they were handled by the corresponding preprocessing pipeline. The same preprocessing logic was applied consistently to the perturbed train and test partitions.

For the high-cardinality scenario, we added two synthetic categorical variables to both the training and test sets. Each variable contained 150 randomly generated categories and was not intended to carry predictive signal. This setting tests whether models can ignore random nuisance categorical structure, such as ID-like sparse fields. It does not evaluate the case where high-cardinality variables contain genuine predictive information.

For both scenarios, models were retrained on the perturbed training data using the best hyperparameters selected during the clean-data HPO stage. We did not retune models specifically for the perturbed settings. This design tests whether clean-selected configurations remain stable when the feature space becomes incomplete or structurally more difficult. Robustness was assessed using predictive performance, training time, model size, and, where available, SHAP-based feature-overlap measures.

III-E. Covariate Shift and SHAP Stability Protocol

We evaluated covariate-shift behavior on the three classification datasets: Titanic, Telco, and DryBean. These datasets were selected because they include binary and multiclass tasks with interpretable tabular features. Tabularized MNIST was excluded because its pixel-derived features are less suitable for semantic feature-attribution analysis, and the Diamonds datasets were excluded because this stage focused on classification metrics together with SHAP-based stability.

Models were trained on the original training data and evaluated on both the clean test set and a shifted copy of the same test set. Only input features were perturbed; labels were kept unchanged. Thus, the protocol approximates covariate shift rather than concept drift.

For numerical features, the shifted test set applied a mean shift of 0.2σ, additive Gaussian noise with standard deviation 0.5σ, and 2% outlier perturbations of magnitude 3σ, where σ is the feature-specific standard deviation estimated from the training data. For categorical features, 10% of values were replaced with the most frequent training category and 5% with a previously unseen category. These magnitudes were chosen as controlled diagnostic perturbations, not as domain-calibrated deployment simulations.

Shift robustness was evaluated using both predictive and attribution-based criteria. Predictive stability was measured by comparing clean and shifted Accuracy and F1-score. Feature-attribution stability was measured using SHAP-based global feature importance, computed as the mean absolute SHAP value per feature. We compared clean and shifted importance profiles using Spearman rank correlation and top-10 Jaccard overlap. Spearman captures broad rank-order stability, while Jaccard captures whether the same dominant predictors remain in the top feature set.

SHAP stability was computed only where reliable explanations were feasible. It was not reported for APT and TabPFN because these pretrained models were evaluated through fixed prediction interfaces that did not allow the same practical and consistent SHAP computation across datasets. Therefore, SHAP-based conclusions are restricted to models with available attribution results. We use SHAP as a relative diagnostic of attribution consistency, not as evidence of causal feature use.

IV. Hyperparameter Optimization Results

IV-A. Cross-dataset performance comparison

We first compare model performance after hyperparameter optimization across the full benchmark. Because the study includes both classification and regression tasks, raw metric values are not directly comparable across datasets. We therefore aggregate results using within-dataset ranks, wins, top-2 frequency, and average gap to the best model. Table III shows that the strongest boosted tree methods occupy the most favorable region of the benchmark. XGBoost, CatBoost, and LightGBM achieve low overall mean ranks and small average gaps to the best model. This indicates that their performance is not driven by a single favorable dataset, but appears repeatedly across the selected tasks. At the same time, these results should be interpreted within the scope of the benchmark and the bounded HPO protocol. The benchmark contains six datasets, and the 30-trial HPO budget is a practical search budget rather than an exhaustive optimization procedure. Thus, the results support a recurring empirical pattern in favor of boosted trees, but they should not be read as a definitive ranking of all tabular model families. The neural models show a more variable profile. MLP and TabNet are competitive on selected datasets, but their overall placement is less stable across the benchmark. The pretrained models provide an additional contrast. TabPFN reduces part of the gap to tuned neural baselines in some classification settings, whereas APT is weaker in this benchmark. Since TabPFN and APT are evaluated in a low-tuning pretrained setting and only on classification tasks, their results are not directly symmetric with the fully tuned models.

IV-B. Classification and regression performance patterns

Across the classification datasets, tree-based methods remain the most consistently competitive group in this study. Their advantage is most visible on smaller or mixed-type datasets, where strong inductive bias and selective feature usage may be beneficial. However, the results are not uniform across all datasets, and the gap narrows in settings with more homogeneous or numerical feature structure. This variation is important because it avoids a simplistic conclusion that neural models always underperform on tabular data. Instead, the benchmark suggests that neural models can be competitive in selected settings, but their strength is more dataset-dependent under the present protocol. The regression results provide an additional structural comparison through *diamonds_v2* and *diamonds_v3*. Both variants come from the same source dataset but use different correlation-filtering thresholds. Tree-based methods remain comparatively stable across these variants, while neural models are more affected by the stronger removal of correlated predictors. This pattern is consistent with the idea

Model	Mean Rank (Cls.)	Mean Rank (Reg.)	Mean Rank (Overall)	Wins	Top-2	Avg. Gap to Best (%)	Time Rank	Size Rank
xgboost	2.25	2.00	**2.17**	1	**4**	6.09	2.83	3.50
catboost	2.00	3.50	2.50	2	3	5.40	4.67	**2.00**
lgbm	**1.75**	4.50	2.67	2	3	**4.80**	3.00	2.83
rf	4.00	**1.00**	3.00	**3**	3	18.78	3.67	4.83
tabnet	3.50	5.50	4.17	2	2	8.83	7.00	5.00
mlp	6.00	4.50	5.50	0	0	13.29	5.17	3.00
tabpfn	5.75	–	5.75	0	0	16.04	5.50	6.75
apt	7.75	–	7.75	0	0	40.11	**1.50**	8.00

TABLE III: HPO model summary across datasets. Ranks are computed within each dataset first, then averaged across datasets. Lower is better for rank/gap/time/size; higher is better for Wins and Top-2. Avg. Gap to Best (%) is computed per dataset and then averaged per model. For each dataset, we take the best score among all models (F1 for classification, R^2 for regression) as the reference, compute each model's percentage shortfall from that reference, and then average these gaps across datasets. Thus, lower values indicate better predictive quality, and 0% means the model matches the dataset-best score.

that neural models may benefit more from redundant support features, whereas tree ensembles can often rely on a smaller set of informative splits.

IV-C. Predictive quality versus computational and model-size trade-offs

Predictive performance alone does not fully determine model suitability for tabular learning. Figure 1 therefore summarizes the average training time and model size across datasets.

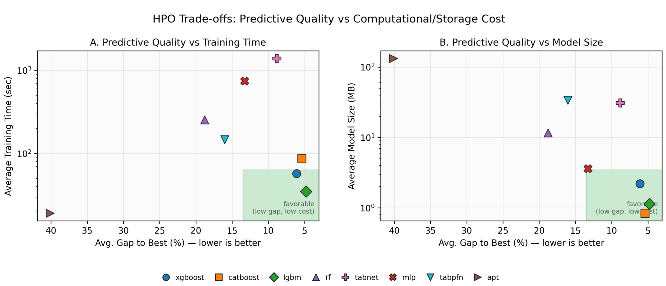

Fig. 1: Predictive quality versus computational and storage cost after hyperparameter optimization. Panel A compares average gap to the best model with training time, and Panel B compares average gap to the best model with model size. Models closer to the lower-right region combine stronger predictive quality with lower cost.

The results indicate that stronger performance does not always require disproportionate computational cost, but neither is there a simple monotonic relation between accuracy and efficiency.

A practically important pattern is that the strongest tree-based methods often sit near a favorable performance–efficiency frontier. In this benchmark, several boosted tree models combine near-best predictive quality with moderate training time and model size. This makes them attractive practical baselines under the present evaluation protocol.: they are not only strong in score, but also competitive in training cost and storage requirements. Neural models, in contrast, may require greater computational burden or larger effective representations while still exhibiting less stable benchmark-wide ranking. This does not make them uncompetitive, but it does reduce their attractiveness when predictive gains are small or inconsistent. The pretrained models occupy an intermediate position: they reduce dataset-specific tuning burden, but this convenience does not automatically translate into superior overall predictive or efficiency trade-offs.

IV-D. Statistical ranking across classification datasets

To complement the raw classification results, we compared model ranks across the four classification datasets using a Friedman test followed by Nemenyi post-hoc analysis at $\alpha = 0.05$. The resulting critical-difference diagram should be interpreted cautiously: its main message is not a strict total ordering of models, but the extent to which observed rank differences are statistically distinguishable.

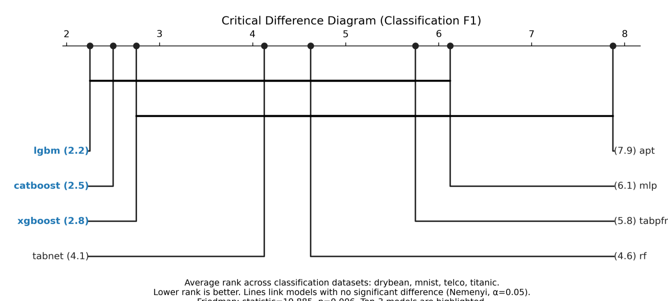

Fig. 2: Critical difference diagram for classification datasets based on F1-score ranks.

In this benchmark, several pairwise differences are not statistically significant, even when the mean-rank gaps appear visually large. This is plausible for two reasons. First, the classification suite is small, containing only four datasets, which limits statistical power. Second, the Nemenyi procedure is conservative, especially when many models are compared simultaneously. In our case, the combination of eight models, a small number of datasets, and several tied or extreme outcomes increases the critical difference and makes broad non-significance unsurprising.

TABLE IV: Model-level robustness summary under missingness and high-cardinality stress. Values are averaged over the available robustness datasets. Performance deltas are measured relative to the clean condition, so values closer to zero indicate greater predictive robustness; positive values indicate improvement under perturbation. Training-time change is reported as percentage change relative to clean training time. SHAP overlap is the average number of shared features in the top-10 feature sets between clean and perturbed conditions, and Jaccard is the corresponding top-10 set overlap ratio. XGBoost is excluded from this section because it was not part of the final robustness comparison. 'NA' denotes unavailable values, primarily because SHAP-based stability quantities were not computed for APT and TabPFN. Mean performance and training-time deltas are taken from the figure-level robustness summaries, while mean Jaccard values are averaged from the available SHAP-delta summary entries.

Model	△Perf. M	△Perf. H	△Time% M	△Time% H	Overlap M	Overlap H	Jaccard M	Jaccard H
CatBoost	-0.0119	-0.0108	-12.23	1.11	9.00	9.25	0.826	0.871
LightGBM	-0.0004	-0.0088	1.97	2.12	9.00	8.50	0.818	0.742
Random Forest	-0.0025	-0.0098	0.43	15.44	9.00	8.25	0.839	0.710
MLP	-0.0077	-0.0194	-7.88	493.18	7.00	5.00	0.617	0.361
TabNet	-0.0002	0.0225	-39.68	-46.62	6.25	5.50	0.537	0.422
APT	0.1200	0.0800	NA	NA	NA	NA	NA	NA
TabPFN	-0.0200	-0.3000	NA	NA	NA	NA	NA	NA

M = missingness, H = high-cardinality.

The diagram therefore supports a restrained conclusion. LightGBM and CatBoost occupy the strongest part of the ranking, while APT lies clearly on the weak end. Beyond those broad separations, however, many intermediate pairwise differences should not be over-interpreted. Thus, the rank-based analysis is most useful here as evidence that the strongest tree-based models remain concentrated near the top of the classification benchmark, rather than as proof of a finely resolved hierarchy among all methods.

V. Robustness to Missingness and High-Cardinality Features

V-A. Robustness under Missingness

The missingness results indicate that tree-based models are comparatively stable when part of the input space is made incomplete. As shown in Table IV, CatBoost, LightGBM, and Random Forest show small average predictive changes and high SHAP top-10 overlap under the missingness stress condition. This suggests that, within the evaluated datasets, these models often continue to rely on similar dominant predictors after moderate input degradation.

CatBoost shows the strongest combined profile in this setting, with a small mean predictive degradation and high feature-overlap stability. LightGBM and Random Forest follow a similar pattern, although their attribution-retention values are slightly lower. The neural models are more mixed. MLP shows a larger predictive drop and lower top-feature stability, while TabNet retains predictive performance more strongly but shows weaker attribution stability. This illustrates an important point: predictive stability and feature-attribution stability are related but not equivalent.

Training-time changes under missingness should be interpreted cautiously. A reduction in training time does not necessarily imply greater robustness; it may also reflect changes in the effective optimization problem after missing-value handling. Therefore, the missingness results are best read jointly across predictive performance, attribution stability, and computational behavior.

V-B. Robustness under High-Cardinality Categorical Stress

The high-cardinality stress scenario produces a clearer separation between model families. In Table IV, tree-based models, especially CatBoost, remain relatively stable after the addition of two random 150-level categorical variables. CatBoost shows only a small mean predictive change and retains high SHAP overlap, while LightGBM and Random Forest also remain comparatively stable.

The neural models are affected more strongly. MLP shows both weaker feature-stability values and a large increase in training time under high-cardinality stress, suggesting that the added nuisance variables increase the representation and optimization burden. TabNet performs better than MLP in predictive terms, but its top-feature stability remains lower than that of the strongest tree models.

This experiment should be interpreted specifically as a nuisance-feature stress test. The added categorical variables are random and intentionally uninformative, so the result mainly indicates how well models ignore irrelevant high-cardinality structure. It does not show how the models would behave when high-cardinality variables contain genuine predictive signal. Under this controlled setting, however, the tree-based models appear more effective at preserving both predictive behavior and dominant feature reliance.

VI. Feature Importance and Stability under Covariate Shift

Covariate shift evaluates a different form of robustness from the missingness and high-cardinality experiments. Here, models are trained on the original training data and evaluated on a shifted test distribution while the labels remain unchanged.

TABLE V: Average predictive and SHAP-based stability under covariate shift. Values are averaged over the three shift datasets. NA indicates unavailable values because the corresponding predictive or SHAP-based quantities could not be computed consistently within the available implementation and runtime environment.

Model	Acc. Clean	Acc. Shift	ΔAcc.	F1 Clean	F1 Shift	ΔF1	SHAP ρ	SHAP Jac.	n_{perf} / n_{SHAP}
CatBoost	0.932998	0.715998	-0.217000	0.864263	0.621144	-0.243119	0.992377	0.939394	3 / 3
RF	0.850254	0.704469	-0.145785	0.620262	0.474098	-0.146163	0.988502	0.878788	3 / 3
MLP	0.747757	0.676971	-0.070786	0.726536	0.636836	-0.089700	0.942586	0.698413	3 / 3
TabNet	0.818573	0.695785	-0.122788	0.843091	0.671167	-0.171924	0.947760	0.467172	3 / 3
TabPFN	0.798919	0.626937	-0.171982	0.787599	0.562084	-0.225515	NA	NA	3 / 0
APT	0.637568	0.593243	-0.044324	0.481393	0.402575	-0.078818	NA	NA	3 / 0
LGBM	NA	NA	NA	NA	NA	NA	NA	NA	0 / 0

Δ values are computed as shifted minus clean, so values closer to zero indicate greater predictive stability. SHAP ρ denotes mean Spearman rank correlation and SHAP Jac. denotes mean Jaccard overlap of the top-10 features. n_{perf} and n_{SHAP} indicate the number of datasets contributing to the predictive and SHAP aggregates, respectively.

The goal is to examine whether predictive performance and feature-importance patterns remain stable when the feature distribution changes. Table V reports the average predictive and SHAP-based stability results across the three shift datasets.

VI-A. Predictive Stability under Shift

The shift results show that clean-data strength and shifted-data stability are related, but not identical. CatBoost retains the strongest average shifted performance in terms of Accuracy and F1-score, even though its average performance drop is not always the smallest. This distinction matters because a small degradation can occur simply when a model starts from a weaker clean baseline.

For this reason, shifted performance should be interpreted jointly with the clean baseline and the magnitude of degradation. Under this view, CatBoost and Random Forest show the most convincing predictive behavior among the models evaluated in this stage. MLP and TabNet remain competitive in selected cases, but their shifted performance and degradation patterns are less consistent across datasets. The pretrained models provide useful reference points, but their comparison is limited because they are classification-only and were not evaluated with SHAP stability.

VI-B. Feature-Importance Stability under Shift

The SHAP analysis asks whether models continue to emphasize similar predictors after the test distribution is shifted. CatBoost and Random Forest show the highest average SHAP stability, with strong Spearman rank correlation and higher top-10 Jaccard overlap than the neural baselines. This suggests that, in the evaluated shift settings, these tree-based models preserve a more stable ordering and top-feature set.

The neural models show a more mixed pattern. MLP retains moderate feature-importance stability, while TabNet shows weaker top-feature retention despite relatively strong predictive performance in some cases. This illustrates why Spearman and Jaccard should be read together: a model may preserve a broadly similar feature ranking while still changing which predictors appear among the most important features.

SHAP stability is used here only as a relative diagnostic of attribution consistency. It does not prove causal feature use, nor does it guarantee predictive robustness. A model can preserve similar feature rankings and still lose performance if the shifted distribution weakens the usefulness of those features. Therefore, the main conclusion from this section is cautious: within the models and datasets where SHAP was available, the strongest tree-based models preserve both predictive performance and attribution structure more consistently than the neural baselines.

VII. Cross-Experiment Analysis

VII-A. Rank Stability versus Robustness

The combined experiments show that clean-data performance is informative, but incomplete. Models that rank highly after HPO often remain competitive under robustness stress and covariate shift, but the relationship is not exact. A strong clean benchmark score does not guarantee robustness under missingness, nuisance categorical structure, or shifted test distributions.

This is the main purpose of Figure 3. It compares predictive retention with SHAP top-10 stability across perturbation settings. Models closer to the region of small predictive change and high attribution stability are more robust under the combined criterion used here. In this benchmark, the strongest tree-based models more often occupy this favorable region, whereas neural models show more variable behavior across stress types.

This result should be interpreted as a cross-experiment pattern rather than a universal ranking. The number of datasets is limited, and not all models have complete SHAP results. Nevertheless, the combined view supports the central point of the study: evaluating tabular models only by clean-data performance can hide important differences in robustness and feature-stability behavior.

VII-B. Performance Drop versus Attribution Drift

Predictive degradation and attribution drift measure different aspects of robustness. Predictive metrics show whether the model still performs well after the data changes. Attribution

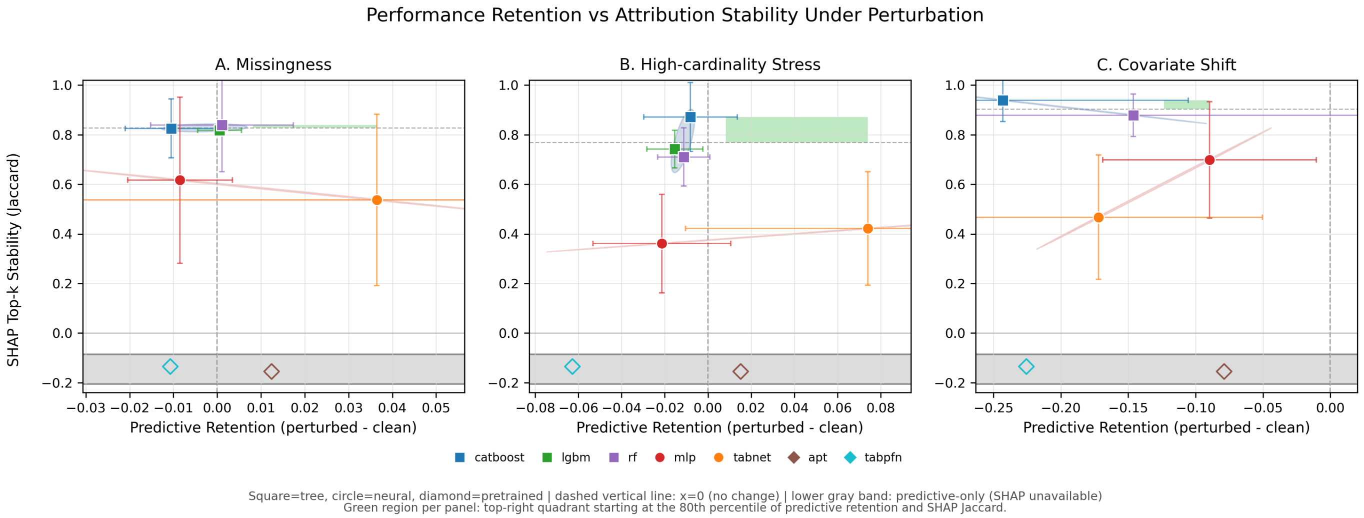

Fig. 3: Performance retention versus attribution stability under three perturbation types: missingness, high-cardinality stress, and covariate shift. Each point denotes a model. The x-axis reports predictive retention relative to the clean condition, and the y-axis reports SHAP top-k stability measured by Jaccard overlap. Error bars indicate dataset-level standard deviation where available. The lower shaded band marks models for which predictive results are available but SHAP-based stability values were not computed. Translucent regions indicate broad model-family groupings, stronger robustness corresponds to smaller predictive change (x-axis values closer to zero) together with higher attribution stability. The green shaded region marks the panel-wise favorable region, defined by high predictive retention and high SHAP Jaccard stability.

stability asks whether the model continues to rely on similar features. These quantities are related, but one should not be treated as a substitute for the other.

This distinction matters because two models may achieve similar shifted performance while relying on different feature sets. Conversely, a model may preserve its main feature ranking but still lose accuracy if the shifted distribution reduces the usefulness of those features. Therefore, the most convincing robustness pattern is not simply low predictive degradation, but the combination of retained predictive performance and stable dominant feature reliance.

Under this combined interpretation, the strongest tree-based models show the most consistent behavior in the present benchmark. This does not prove that tree models are universally more robust, but it suggests that their clean-data strength is supported by more stable behavior under the controlled stress conditions studied here.

VIII. Why Tree-Based Models Remain Strong on Tabular Data

VIII-A. Selective Feature Usage under Nuisance Structure

One explanation for the observed results is that tree-based models are more selective in how they use features. In tree ensembles, a feature affects the model only if it is selected for a split that improves the training objective. Irrelevant or weakly useful features can therefore remain unused. This property is helpful under the high-cardinality stress test, where the added categorical variables are random nuisance features rather than predictive signals.

The empirical results are consistent with this interpretation. Under high-cardinality stress, CatBoost shows only a small mean performance change and preserves high SHAP top-10 overlap, while MLP shows weaker feature stability and a large increase in training time. This suggests that the neural model is more affected by the expanded encoded feature space, whereas the strongest tree models better suppress the nuisance structure. This does not imply that trees always handle high-cardinality variables better; if such variables contain real signal, the comparison may change.

VIII-B. Stable Reliance on Core Predictors

A second explanation is that the strongest tree-based models preserve a more stable notion of which features matter. Across missingness, high-cardinality stress, and covariate shift, CatBoost and Random Forest generally retain higher SHAP stability than the neural baselines. In the covariate-shift analysis, for example, the tree-based models show higher average Spearman rank correlation and top-10 Jaccard overlap than MLP and TabNet.

This matters because robust prediction and stable explanation are not identical. A model may keep acceptable predictive performance while changing which features it emphasizes. In applications where feature semantics matter, such attribution drift can reduce trust in the model. The results therefore suggest that the tree-based advantage in this benchmark is not only predictive, but also linked to more stable reliance on dominant predictors.

VIII-C. Tabular Inductive Bias

A third explanation is the match between tree-based inductive bias and common tabular structure. Many tabular datasets contain heterogeneous feature types, sparse informative variables, threshold-like effects, local interactions, and irregular decision boundaries. Split-based models operate directly on the original feature space and can exploit such structure through local partitions.

This interpretation is consistent with prior work arguing that tree-based models remain strong on typical tabular data because they handle uninformative features, preserve feature orientation, and learn irregular functions effectively [11]. Our experiments do not prove this mechanism directly, but they are consistent with it: tree-based models remain comparatively stable when the feature space is made incomplete, expanded with nuisance categorical variables, or shifted at test time.

VIII-D. Neural and Pretrained Models

The results do not imply that neural or pretrained tabular models are generally weak. MLP, TabNet, TabPFN, and APT are competitive in selected settings. However, in this benchmark they are less consistent across clean performance, perturbation robustness, attribution stability, and efficiency.

For the neural baselines, the main issue appears to be sensitivity to feature-space changes and tuning conditions rather than lack of capacity alone. For the pretrained models, the main advantage is reduced dataset-specific tuning, but they do not establish a new robust frontier in the evaluated settings. Because APT and TabPFN were not tuned in the same way as the other models and did not have SHAP-stability results, conclusions about pretrained models should remain limited.

IX. Discussion and Conclusion

IX-A. Limitations and Future Prospects

Although this study combines clean performance, robustness stress tests, covariate shift, and feature-stability analysis, several limitations remain.

First, the benchmark is moderate in scale. It includes different task types and feature structures, but only six datasets overall; the robustness analysis uses four datasets and the covariate-shift analysis uses three. Therefore, the findings should be interpreted as recurring empirical patterns rather than universal claims about tabular learning.

Second, the perturbation protocols are controlled heuristic stress tests. Missingness, high-cardinality categorical noise, and covariate shift were designed to isolate specific failure modes, but they are not domain-calibrated simulations of real deployment environments. Natural temporal shifts, domain-specific missingness mechanisms, and predictive high-cardinality identifiers may lead to different outcomes.

Third, the HPO protocol is bounded. Using the same Optuna trial budget gives a simple comparison protocol, but equal trial counts do not guarantee equal effective tuning effort across model families. Tree ensembles, neural networks, and pretrained models have different search-space sizes and tuning sensitivities. Larger budgets, repeated tuning runs, and more extensive seed analysis could change some rankings.

Fourth, uncertainty quantification is limited. The study reports aggregate ranks, gaps, perturbation deltas, and SHAP-stability measures, but does not fully include repeated cross-validation, confidence intervals, bootstrap uncertainty, or formal significance testing for all robustness metrics. The Friedman/Nemenyi comparison is also limited by the small number of classification datasets.

Fifth, dataset-size effects are not studied systematically. The selected datasets vary in sample size, but the paper does not include controlled learning curves or subsampling experiments. This is especially relevant for neural tabular models such as TabNet, which may require more data to show their representational advantages.

Sixth, model coverage is incomplete in some evaluation stages. LightGBM could not be fully evaluated in the covariate-shift setting, and SHAP-based stability was unavailable for APT and TabPFN. In addition, the pretrained models were evaluated in their default or fixed-checkpoint setting and were classification-only in this study. Conclusions about pretrained models should therefore remain limited.

Finally, SHAP-based stability should be interpreted carefully. It measures whether attribution patterns remain similar under perturbation or shift, but it does not establish causal feature use or complete interpretability. Future work should extend the benchmark scale, add learning-curve and train–test gap analysis, include stronger uncertainty estimates, evaluate additional modern neural baselines such as FT-Transformer or ResNet-style tabular models, and study probabilistic quality, calibration, uncertainty estimation, and the usefulness of neural or pretrained embeddings.

IX-B. Conclusion

This paper examined tabular model behavior beyond clean benchmark ranking by combining hyperparameter-optimized evaluation with robustness stress tests and feature-stability analysis. Within the selected benchmark, tree-based models, especially boosted tree ensembles such as CatBoost, LightGBM, and XGBoost, showed the most consistent empirical pattern across clean performance, controlled perturbations, covariate shift, and attribution stability.

The results suggest that the continued strength of tree-based models in this study is linked not only to predictive accuracy, but also to their ability to preserve useful behavior under controlled feature-space stress. Neural and pretrained tabular models remain competitive in selected cases, but their performance and stability were less consistent under the combined evaluation criteria used here.

These conclusions should not be read as a universal proof of tree-model dominance. Rather, the study supports a more careful evaluation message: tabular models should be compared not only by clean test accuracy, but also by robustness, feature-attribution stability, computational behavior, and the limits of the evaluation protocol itself.

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